

Exercise 2: Stored Procedures

Aim

To create and execute Stored Procedures in MySQL for retrieving and inserting employee details.

Theory

What is a Stored Procedure?

A **Stored Procedure** is a collection of SQL statements stored in the database that can be executed whenever required. It helps automate repetitive database operations.

Advantages of Stored Procedures

- Improves query performance.
- Reduces code duplication.
- Enhances database security.
- Makes applications easier to maintain.
- Allows parameter passing for dynamic queries.

Parameters in Stored Procedures

- **IN** – Accepts input values.
- **OUT** – Returns values to the caller.
- **INOUT** – Works as both input and output.

Syntax

```
CREATE PROCEDURE ProcedureName()
```

```
BEGIN
```

```
    SQL Statements;
```

```
END;
```

Applications

- Inserting records.
- Retrieving records.
- Updating records.
- Deleting records.

- Generating reports.

Conclusion

Stored Procedures simplify database operations by storing SQL logic inside the database. They improve performance, security, and code reusability.

CODE :

Create the Employees Table

```
1 • CREATE DATABASE EmployeeDB;
2 • USE EmployeeDB;
3
4 • CREATE TABLE Employees(
5     EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
6     FirstName VARCHAR(50),
7     LastName VARCHAR(50),
8     DepartmentID INT,
9     Salary DECIMAL(10,2),
10    JoinDate DATE
11 );
12 • SHOW DATABASES;
13
```

Insert Sample Data:

- `INSERT INTO Employees`
(FirstName, LastName, DepartmentID, Salary, JoinDate)
`VALUES`
(`'Ravi', 'Kumar', 101, 50000, '2023-01-15'`),
(`'Priya', 'Sharma', 102, 60000, '2022-05-10'`),
(`'Rahul', 'Reddy', 101, 55000, '2024-02-20'`);
- `SELECT * FROM Employees;`

Output :

Result Grid						
Filter Rows:						
Edit: Export/Import: Wrap Cell Content:						
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
▶	1	Ravi	Kumar	101	50000.00	2023-01-15
	2	Priya	Sharma	102	60000.00	2022-05-10
	3	Rahul	Reddy	101	55000.00	2024-02-20
	4	Anitha	Rao	103	65000.00	2025-06-20
•	NULL	NULL	NULL	NULL	NULL	NULL

Create a Stored Procedure to Retrieve Employees by Department :

```
--  
3 DELIMITER //  
4  
5 • CREATE PROCEDURE GetEmployeeByDepartment(IN DeptID INT)  
6 BEGIN  
7     SELECT *  
8     FROM Employees  
9     WHERE DepartmentID = DeptID;  
10 END //  
11  
12 DELIMITER ;  
13
```

Execute the Stored Procedure :

```
CALL GetEmployeeByDepartment(101);
```

```
DELIMITER //
```

OUTPUT :

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
▶	1	Ravi	Kumar	101	50000.00	2023-01-15
	3	Rahul	Reddy	101	55000.00	2024-02-20

Create the sp_InsertEmployee Stored Procedure :

```
7  
8 • CREATE PROCEDURE sp_InsertEmployee(  
9     IN First_Name VARCHAR(50),  
10    IN Last_Name VARCHAR(50),  
11    IN Department_ID INT,  
12    IN Salary_Value DECIMAL(10,2),  
13    IN Join_Date DATE  
14 )  
15 BEGIN  
16     INSERT INTO Employees  
17     (FirstName, LastName, DepartmentID, Salary, JoinDate)  
18     VALUES  
19     (First_Name, Last_Name, Department_ID, Salary_Value, Join_Date);  
20 END //  
21  
22 DELIMITER ;  
23
```

Execute the Insert Procedure :

```
CALL sp_InsertEmployee(  
    'Anitha',  
    'Rao',  
    103,  
    65000,  
    '2025-06-20'  
);  
  
SELECT * FROM Employees;
```

Output :

Result Grid						
Filter Rows:						
Edit: Export/Import: Wrap Cell Content:						
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
▶	1	Ravi	Kumar	101	50000.00	2023-01-15
	2	Priya	Sharma	102	60000.00	2022-05-10
	3	Rahul	Reddy	101	55000.00	2024-02-20
	4	Anitha	Rao	103	65000.00	2025-06-20
	5	Anitha	Rao	103	65000.00	2025-06-20
•	NULL	NULL	NULL	103	NULL	NULL